Revival: Collaborative Artistic Creation through Human-AI Interactions in Musical Creativity

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1 Introduction

Artwork concept: Revival is an artwork and innovative live audiovisual performance by the artist collective K-Phi-A, which integrates human and machine musicianship through artificial intelligence to produce electronic music and its audio-reactive visuals. The performance is centred on real-time co-creative improvisation between human performers—a percussionist and an electronic music performer—and AI musical agents, such as MASOM [1] and SpireMuse [2]. These musical agents are trained on the works of deceased composers and compositions created by the collective, enabling them to respond dynamically to human input and emulate complex musical styles. An AI-driven visual synthesizer trained on public domain images, Autolume [3], generates visuals responsive to the evolving musical landscape with a human VJ. This performance exemplifies the potential of AI in co-creative artistic creation, demonstrating how AI and human artists can collaboratively create in a live music improvisation context. The comprehensive performance setup is detailed in Appendix A.

Artist collective: The K-Phi-A consists of percussionist Keon Ju Maverick Lee, who collaborates with musical agents; electronic music performer Philippe Pasquier, who utilizes a DIY software sampler named Kenaxis and agents; and VJ Amagi, who operates the Autolume audiovisual synthesizer.

Dr. Philippe Pasquier is a scientific researcher, a multidisciplinary media artist, an educator, and a community builder. He is the director of the Metacreation Lab for Creative AI and a Professor at Simon Fraser University (SFU), School for Interactive Arts and Technology in Vancouver, Canada. He is pursuing a multidisciplinary research-creation program focused on generative systems and applied AI for creative tasks there. His music AI algorithms have reached millions through collaborations with the software industry (e.g., Steinberg, Ableton, Teenage Engineering, Elias, etc.), and his artworks have been shown on six continents.

Keon Ju Maverick Lee is an electronic percussionist, drummer, music composer, and sound artist focusing on research-creation for designing a musical agent and interactive music systems to accompany drum performances. He has been active as a drummer in the local music scene in British Columbia, Canada. Keon is a Ph.D. student in the Metacreation Lab for Creative AI at Simon Fraser University. He is a lab instructor for the Sound Design course at SFU and computer music courses at the University of Victoria. As an academic, he publishes research-creation papers in music and AI. He collaborates with colleagues at the Centre for Digital Music in London, UK, and IRCAM in Paris, France. Recently, he has been collaborating with a media artist to get involved in the ReVerie project performed at Ars Electronica and SIGGRAPH [4] in 2024.

VJ Amagi is a VJ (Video Jockey) / graphics programmer based in Vancouver, Canada. He performs with real-time visuals and creates art installations using TouchDesigner and Unity. Additionally, he develops VJ systems for shader live coding performances. At MUTEK.jp 2018, he led a workshop on visual effects using TouchDesigner.

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2 Human-Machine Interaction for Music Improvisation with Audiovisual

The core element of our human-machine interaction framework is the machine listening module [5] embedded within the AI musical agent system [6]. This module integrates with our conductor software environment, utilizing Chataigne, AbletonLive, and the AI-driven visual synthesizer Autolume. The artistic exploration examines the possibilities of musical practices facilitated by a co-creative framework [11], employing a research-creation methodology [12] to investigate the dynamic interplay between music and reactive audiovisuals as a medium. This inquiry operates at the nexus of music improvisation, audiovisual art, and media art, emphasizing human-AI collaboration in co-creation. Our approach advocates for a small data mindset [10], highlighting the responsible and ethical utilization of data resources. Small data offers ethical advantages by enhancing transparency around copyrighted material for artists and musicians. Focusing on curated datasets limits excessive data scraping and promotes responsible resource use, aligning with principles of data minimalism and artist protection. Furthermore, adopting a small data mindset is more environmentally sustainable, demanding significantly fewer computational resources for training processes than big data systems.

Offline machine listening: Musical agents engage with a percussionist and electronic music performances within our interactive music system, leveraging custom machine listening algorithms. This system employs both low-level audio features and high-level affective features to enhance the interactivity of the musical agents. Each audio segment is annotated with a feature vector consisting of 55 dimensions, which includes duration, mean and standard deviation of loudness, Mel-Frequency Cepstral Coefficients (MFCC), fundamental frequency, chroma, and high-level affective features such as valence and arousal [1]. To achieve more precise melodic and harmonic analysis, a large FFT (Fast Fourier Transform [7]) window size of 8192 samples combined with a smaller hop size of 512 samples was employed, providing high-frequency resolution. This configuration was chosen based on experimental results. We incorporate chroma features, which represent pitch histograms corresponding to the 12 chromatic scale notes, significantly enhancing the musical agents' ability to detect harmonic and melodic characteristics. Chroma features enhance melodic classification and reduce pitch errors in audio segments with multiple notes, providing a discrete representation that facilitates the identification of harmonic content. This approach also improves clustering within the self-organizing map [8], even when the calculated pitch inaccurately reflects tonality. In our affective computing module of the agent, Valence and arousal, as defined by the circumplex model of affect [9], are computed from a multivariate linear regression with coefficients of low-level audio features and further enhance the system's capacity to recognize the emotional characteristics of sound samples.

Real-time machine listening: During runtime, the machine listening algorithm continuously segments the user's input stream into segments aligned with those in the reference corpus. The listening module can adjust feature weights, subsequently influencing the matching algorithms. In SpireMuse, four influence dimensions—rhythmic, spectral, melodic, and harmonic—are available as parameters [2]. The rhythmic parameter prioritizes the duration feature, while the spectral parameter emphasizes Mel-Frequency Cepstral Coefficients (MFCC) features. The melodic parameter focuses on the fundamental frequency, and the harmonic parameter highlights chroma features. MASOM is designed to handle primary musical passages within the musical agent system, whereas SpireMuse accompanies a percussionist and electronic music performer, leveraging its real-time machine listening module.

Audiovisual: In the interactive component of our audiovisual system, spectral and Bark coefficients are extracted from the audio outputs of Keon, Philippe, and the musical agents. These coefficients are transmitted via Open Sound Control (OSC) messages to Autolume and VJ Amagi, enabling the generation of reactive visuals in real-time, which are shaped by VJ Amagi's artistic techniques for visual music. Furthermore, the extracted audio features are conveyed to the Digital Multiplex (DMX) interface to control reactive lighting, facilitating visual representation for each performer. The reactive lighting system enhances the visual interpretation of each performer's actions by colour-mapping specific audio features to the lighting system, thus enriching the audience's performance experience.

Conductor: We utilize Chataigne software to configure our conductor system, which orchestrates communication, sound/visual parameter control, and timelines for our structured live music improvisation and audiovisual performance. The performance comprises multiple musical and visual themes, each employing distinct sound parameters for musical agents, percussion samples, audio samples, and visual elements. The conductor coordinates these elements in real-time through OSC messages to interact with the musical agents, visual synthesizer, and DMX lighting systems.

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